

Supply Chain Logistics Manager - Data Model & Entity Relations

Confidential - Internal Use Only

Version 1.0

1. Core Entities

Entity: Shipment - Fields: shipment_id (UUID PK), customer_id (UUID FK), origin_address_id (UUID FK), dest_address_id (UUID FK), carrier_id (UUID FK NULL), tracking_number (VARCHAR 50 UNIQUE NULL), service_level (ENUM: express, ground, economy, freight), status (ENUM: created, label_printed, picked_up, in_transit, out_for_delivery, delivered, exception, cancelled, returned), total_weight_kg (DECIMAL 10,3), declared_value (DECIMAL 12,2), currency (CHAR 3), estimated_delivery (TIMESTAMPTZ NULL), actual_delivery (TIMESTAMPTZ NULL), created_at (TIMESTAMP).

Entity: Package - Fields: package_id (UUID PK), shipment_id (UUID FK), sequence_number (SMALLINT), weight_kg (DECIMAL 8,3), length_cm (DECIMAL 6,1), width_cm (DECIMAL 6,1), height_cm (DECIMAL 6,1), package_type (ENUM: envelope, box, pallet, tube, custom), tracking_number (VARCHAR 50 NULL).

Entity: Carrier - Fields: carrier_id (UUID PK), name (VARCHAR 100), code (VARCHAR 10 UNIQUE), api_type (ENUM: rest, soap, edi), api_base_url (VARCHAR 500), credentials_encrypted (BYTEA), supported_services (JSONB), active (BOOLEAN DEFAULT true).

Entity: TrackingEvent - Fields: event_id (BIGSERIAL PK), shipment_id (UUID FK), event_type (VARCHAR 50), description (TEXT), location_city (VARCHAR 100), location_country (CHAR 2), latitude (DECIMAL 9,6 NULL), longitude (DECIMAL 9,6 NULL), event_timestamp (TIMESTAMPTZ), carrier_raw_status (VARCHAR 50), created_at (TIMESTAMP).

Entity: InventoryItem - Fields: item_id (UUID PK), warehouse_id (UUID FK), sku (VARCHAR 50), product_name (VARCHAR 255), quantity_on_hand (INTEGER), quantity_reserved (INTEGER), quantity_available (INTEGER GENERATED), bin_location (VARCHAR 20), reorder_point (INTEGER), unit_weight_kg (DECIMAL 8,3), hs_code (VARCHAR 12 NULL), last_counted_at (TIMESTAMP).

Entity: WarehouseOrder - Fields: order_id (UUID PK), warehouse_id (UUID FK), customer_id (UUID FK), order_type (ENUM: outbound, inbound, transfer), status (ENUM: received, wave_planned, picking, packing, staged, shipped, cancelled), priority (ENUM: standard, rush, same_day), ship_by_date (DATE), created_at (TIMESTAMP).

Entity: CustomsDeclaration - Fields: declaration_id (UUID PK), shipment_id (UUID FK), destination_country (CHAR 2), total_declared_value (DECIMAL 12,2), currency (CHAR 3), hs_codes (JSONB), screening_status (ENUM: pending, cleared, flagged), screening_result (JSONB NULL), duties_estimated (DECIMAL 10,2 NULL), filed_at (TIMESTAMP NULL).

2. Relationships

Shipment 1:N Package - A shipment contains one or more packages.

Carrier 1:N Shipment - A carrier handles many shipments.

Shipment 1:N TrackingEvent - A shipment has many tracking events over its lifecycle.

Warehouse 1:N InventoryItem - A warehouse stores many SKU-level inventory items.

WarehouseOrder 1:1 Shipment - Each fulfilled order generates one shipment.

Shipment 1:0..1 CustomsDeclaration - International shipments have customs declarations.

3. Time-Series Data

REQ-LOG-073: TrackingEvent table uses TimescaleDB hypertable partitioned by event_timestamp.

Retention: 2 years hot, 5 years cold (S3 Glacier).

REQ-LOG-074: Sensor readings (temperature, humidity, shock) stored in InfluxDB time-series database with 1-second granularity during transit.

REQ-LOG-075: Carrier rate quotes cached in Redis with 30-minute TTL. Historical rates archived to PostgreSQL for trend analysis.

4. Data Partitioning & Performance

REQ-LOG-076: Shipment table partitioned by created_at month. Current + 3 months in hot storage (SSD). Older in warm storage (HDD).

REQ-LOG-077: InventoryItem table indexed on (warehouse_id, sku) and (warehouse_id, bin_location) for fast lookup during picking.

REQ-LOG-078: Materialized views for analytics: daily_shipment_stats, carrier_performance_30d, warehouse_utilization. Refreshed every 15 minutes.